

Q1. Match the following physical concepts with their corresponding detailed analysis:

1. Concept Analysis

- I. Torque A. Involves comparing the effects of forces around a fixed point  
II. Moment of Inertia B. Involves calculation of distance squared times mass for rotational effects  
III. Radius of Gyration C. Measures how mass is distributed relative to the axis of rotation  
IV. Angular Momentum D. Conserved in a closed system and involves rotational motion
- A. I-A, II-B, III-D, IV-C  
B. I-C, II-D, III-A, IV-B  
C. I-A, II-B, III-C, IV-D  
D. I-B, II-A, III-D, IV-C

Answer: Option 3

Solution: Torque is the rotational equivalent of force and primarily involves understanding the effects around a fixed point (I-A). Moment of Inertia is a measure of an object's resistance to changes in its rotation rate (II-B). Radius of Gyration deals with the distribution of the mass and its distance from the axis, which helps in simplifying the calculation of moments of inertia (III-C). Angular Momentum, similar to linear momentum, is conserved in a closed system and is crucial in the analysis of rotational motions (IV-D). Hence, Option (3) is the right answer.

Q2. Match the following molecular concepts with their correct explanations:

1. Concept Explanation

- I. Molecular-Orbital Theory A. Describes bonding in terms of overlapping atomic orbitals  
II. Hybridization B. Explains molecule shapes using regions of electron density  
III. VSEPR C. Accounts for molecular geometry by electron pair repulsion  
IV. Valence-Bond Theory D. Involves combination of atomic orbitals to form molecular orbitals
- A. I-A, II-B, III-C, IV-D  
B. I-D, II-B, III-C, IV-A  
C. I-B, II-D, III-A, IV-C  
D. I-D, II-A, III-C, IV-B

Answer: Option 4

Solution: Molecular-Orbital Theory describes the behavior of electrons in molecules by assuming that the electrons occupy molecular orbitals that spread throughout the molecule (I-D). Hybridization is the process by which atomic orbitals mix to form new hybrid orbitals that are suitable for pairing electrons to form chemical bonds (II-A). VSEPR is a model used to predict the geometry of molecules based on the electron-pair repulsion between their bonded atoms (III-C). Valence-Bond Theory explains the chemical bond formation between atoms using the concept of overlapping atomic orbitals where electron pairs are shared (IV-A). Hence, Option (4) is the right answer.

Q3. Match the following motion concepts with their relevant detailed analysis:

1. Concept Analysis

- I. Projectile Motion A. Motion combining both horizontal and vertical components  
II. Relative Velocity B. Describes the velocity of an object as observed from another moving object  
III. Uniform Circular Motion C. Involves constant speed but changing velocity due to direction change  
IV. Motion in a Plane D. General term covering more than one dimension of motion including the above
- A. I-D, II-B, III-A, IV-C  
B. I-C, II-B, III-A, IV-D  
C. I-A, II-B, III-C, IV-D  
D. I-B, II-A, III-D, IV-C

Answer: Option 3

Solution: Projectile Motion is a type of motion where an object moves in a curved path under the influence of gravity, involving both horizontal and vertical components (I-A). Relative Velocity is the velocity of one body with respect to another and is crucial when analyzing moving objects from different frames of reference (II-B). Uniform Circular Motion, while maintaining a constant speed, involves a continuous change in direction, which means the velocity is not constant (III-C). Motion in a Plane covers all motions that occur in two dimensions, including projectile and circular motions (IV-D). Hence, Option (3) is the right answer.

Q4. In trigonometric equations, the expression  $\sin^2(x) + \cos^2(x)$  equals \_\_\_\_\_.

- A. 1
- B. 0
- C. x
- D. pi

Answer: Option 1

Solution: This is a fundamental trigonometric identity.  $\sin^2(x) + \cos^2(x)$  always equals 1, regardless of the value of x. Hence, Option (1) is the right answer.

Q5. The shape of a molecule determined by the VSEPR theory depends on the \_\_\_\_\_ of the electron pairs around the central atom.

- A. repulsion
- B. attraction
- C. hybridization
- D. resonance

Answer: Option 2

Solution: VSEPR theory states that the shape of a molecule is mainly determined by the repulsion between the electron pairs surrounding the central atom. Hence, Option (2) is the right answer.

Q6. The moment of inertia is greater when mass is distributed \_\_\_\_\_ from the axis of rotation.

- A. closer
- B. further
- C. equally
- D. randomly

Answer: Option 2

Solution: Moment of inertia increases as the mass moves further from the axis of rotation because it depends on the distance squared from the axis. Hence, Option (2) is the right answer.